# CSC9008 Cyber-Physical Systems
# The good plate: Final Presentation

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# Agenda

1. System Architecture
2. Recall: Basic background
3. Complementary Filter for Attitude Estimation
4. PID Control System
5. String-Length Geometry for MG90S Servos
6. System analysis
7. Live Demo

# Project Overview

## Goal

Build a self-stabilizing platform that keeps objects level despite external disturbances (manual tilting, vibration, earthquakes).

**Hardware:**

- STM32F401CCU6 (MCU)
- GY-521 / MPU6050 (IMU)
- 4 $\times$ MG90S servos
- Pulleys + strings
- Central universal joint

**Control flow:**

- Sense tilt via IMU
- Compute attitude (pitch, roll)
- PID computes correction
- Servos pull strings
- Platform tilts back to level

# Hardware Price

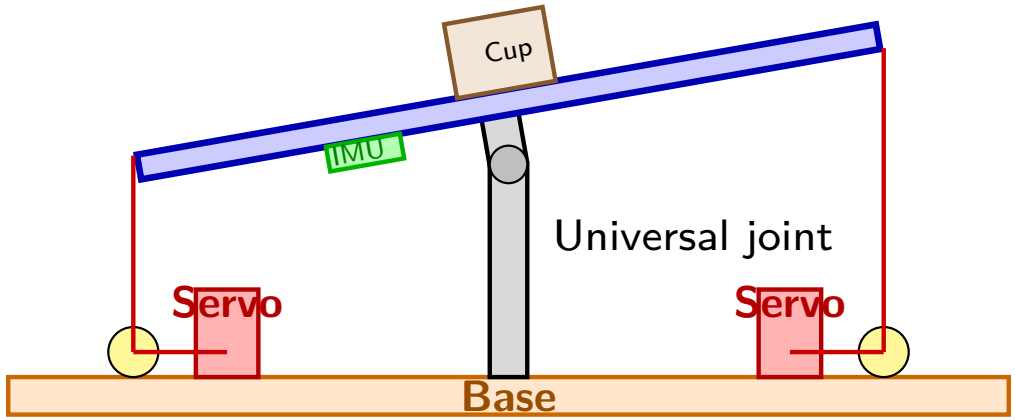
| Component | Unit Price (NTD) | Quantity | Subtotal (NTD) |
|---|---|---|---|
| STM32F401CCU6 | $400 | 1 | $400 |
| GY-521 / MPU6050 | $95 | 1 | $95 |
| MG90S Servo | $50 | 4 | $200 |
| Clipboard | $29 | 1 | $29 |
| Pulley | $19 | 4 | $76 |
| String | $14 | 1 | $14 |
| Universal Joint | $76 | 1 | $76 |
| Socket | $23 | 1 | $23 |
| Screw Eye | $12 | 1 pack | $12 |
| Serrated Flange Nut | $20 | 1 pack | $20 |
| Washer | $10 | 1 pack | $10 |
| Corrugated Board | $12 | 1 | $12 |
| Acrylic Sheet | $140 | 1 | $140 |
| Dupont Wires | $20 | lots | $20 |
| Arduino Mega | ($400) | 1 | ($400) |
| ST-link V2 | ($200) | 1 | ($200) |
| **Total** | | | **$1127(1727)** |

# Quick Demo

**Quick Live Demo**

(We will have a more detailed demo at the end of the presentation)

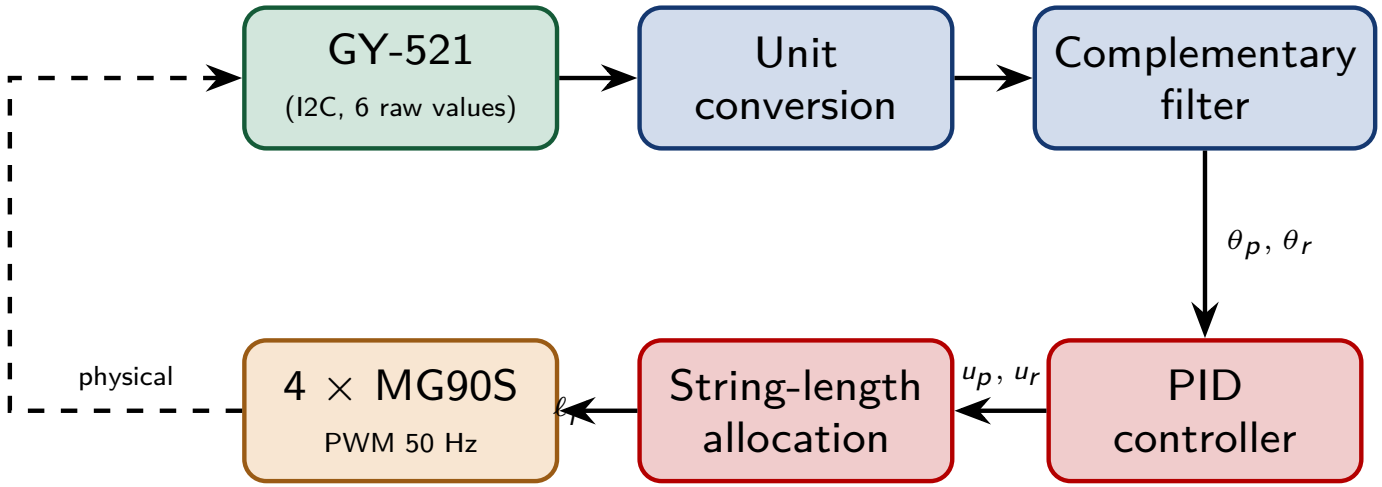
# Mechanical Structure




- 4 strings pulling 4 platform corners (PA0 back, PA1 left, PA2 right, PA3 front)
- Cross-pairs work antagonistically: one pulls in, the other releases
- Central universal joint bears the load; servos only adjust angle

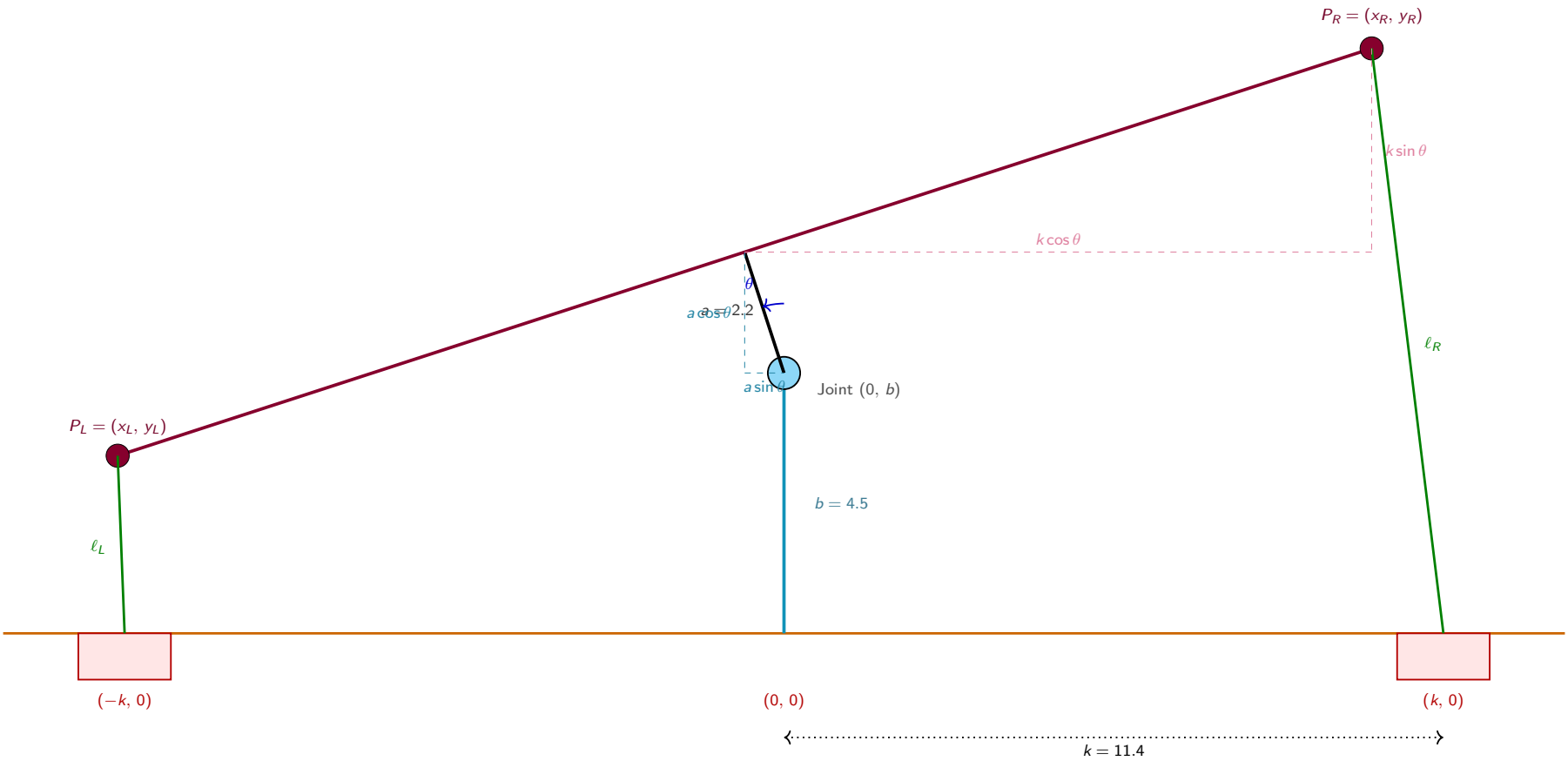
Acknowledgement: Chiin.H().

# Signal Flow Diagram




- Closed-loop at **500 Hz** (2 ms control period)
- Pitch and roll axes controlled independently
- All computation runs on STM32F401, USB-CDC for logging

# Geometric Setup of the Mechanism




Platform tilts about the universal joint at $(0, b)$. The *upper pillar* (length $a$) rigidly connects the joint to the platform center, so it tilts with the platform. Strings run from platform ends to fixed servos at $(\pm k, 0)$.

# Servo Action

Corrections $u_p$, $u_r$ in **degrees of attitude correction**.
But each servo controls the **length of a string** that pulls the platform.

## Naive approach (does not work well)

Map servo angle directly to angle output: $\theta_{\text{servo}} += u$.
Problem: $1^\circ$ of servo rotation does *not* produce $1^\circ$ of platform tilt. The relationship is highly nonlinear because strings connect a rotating platform to fixed servo positions.

## Our approach

Compute the *exact string length* the platform would need at the desired tilt angle, then convert that length into a servo rotation via the spool radius.

# Deriving the String Length (Step 1: Locate Platform Center)

The upper pillar rotates rigidly with the platform about the joint at $(0, b)$. The platform *center* is the top of the upper pillar. Before rotation it sits at $(0, b + a)$. The offset from the joint is $(0, a)$. After rotation by $\theta$:

$$\text{offset}' = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} 0 \\ a \end{bmatrix} = \begin{bmatrix} -a\sin\theta \\ a\cos\theta \end{bmatrix}$$

Add back the joint position $(0, b)$:

$$\boxed{P_{\text{center}} = \left(-a\sin\theta,\ b + a\cos\theta\right)}$$

This term is what makes our model different from a naive single-pillar setup. Because $a$ is not negligible compared to $k$, ignoring it would introduce a position error of up to $a\sin\theta_{\max}$ in the platform center.

# Deriving the String Length (Step 2: Platform Endpoints)

Each platform endpoint is offset $\pm k$ along the platform direction. After rotation, the offsets become $(\pm k\cos\theta,\ \pm k\sin\theta)$.
Add to the platform center $P_{\text{center}}$:

$$P_R = \big(-a\sin\theta + k\cos\theta,\ b + a\cos\theta + k\sin\theta\big)$$

$$P_L = \big(-a\sin\theta - k\cos\theta,\ b + a\cos\theta - k\sin\theta\big)$$

String length is the Euclidean distance from each endpoint to its servo:

$$\ell_R = \|P_R - (k,0)\|, \quad \ell_L = \|P_L - (-k,0)\|$$

## Result

$$\ell_R(\theta) = \sqrt{(-a\sin\theta + k\cos\theta - k)^2 + (b + a\cos\theta + k\sin\theta)^2}$$

$$\ell_L(\theta) = \sqrt{(-a\sin\theta - k\cos\theta + k)^2 + (b + a\cos\theta - k\sin\theta)^2}$$

# Sanity Check at $\theta = 0$

Substituting $\sin 0 = 0$, $\cos 0 = 1$:

$$\ell_R(0) = \sqrt{(0 + k - k)^2 + (b + a + 0)^2} = a + b$$

$$\ell_L(0) = \sqrt{(0 - k + k)^2 + (b + a - 0)^2} = a + b$$

Both strings have length $a + b$ at rest ✓

String length *change* relative to neutral position:

$$\Delta\ell_R(\theta) = \ell_R(\theta) - (a + b), \qquad \Delta\ell_L(\theta) = \ell_L(\theta) - (a + b)$$

## Numerical example: $a = 2.2$, $b = 4.5$, $k = 11.4$, $\theta = 10^\circ$

$$\ell_R(10^\circ) \approx 8.66 \text{ cm}, \quad \Delta\ell_R \approx +1.96 \text{ cm (release)}$$
$$\ell_L(10^\circ) \approx 4.69 \text{ cm}, \quad \Delta\ell_L \approx -2.01 \text{ cm (reel in)}$$

The two strings move *antagonistically* (one shortens, one lengthens) — exactly the behavior we want.

# From String Length to Servo Angle

Each servo has a spool with circumference $C$. One full 360° revolution winds/unwinds one full circumference of string. Therefore:

$$\Delta\ell = C \cdot \frac{\alpha}{360^\circ} \quad \Longrightarrow \quad \boxed{\alpha = \frac{360^\circ \cdot \Delta\ell}{C}}$$

where $\alpha$ is the servo rotation (in degrees) and $\Delta\ell$ is the required string-length change.

Final servo command (relative to the 90° neutral position):

$$\theta_{\text{servo}} = 90^\circ + \alpha$$

## Continuing the example ($\Delta\ell_R = +1.96$ cm, $C = 2\pi r$ with $r = 1.0$ cm $\Rightarrow C \approx 6.28$ cm)

$$\alpha_R = \frac{360 \cdot 1.96}{6.28} \approx 112^\circ$$

A 112° servo rotation is required to release 1.96 cm of string — which is why bigger spool circumference gives finer (but smaller-range) control.

# What the MPU6050 Measures

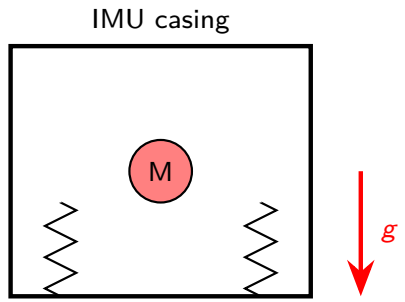
## Two sensors in one chip

**Accelerometer:** measures *specific force* (in g)
**Gyroscope:** measures *angular rate* (in deg/s)

**Accelerometer (counter-intuitive!)**
Even when stationary, the chip reads $1\,g$ on $z$ because:

$$\vec{a}_{\text{meas}} = \vec{a}_{\text{motion}} - \vec{g}$$

The spring inside resists gravity $\Rightarrow$ measurement is the reaction force.



# Fuse Two Sensors

| | **Accelerometer** | **Gyroscope** |
|---|---|---|
| Steady state | Accurate (gravity) | Slow drift |
| Fast motion | Corrupted by inertia | Accurate |
| Long term | No drift | Drift grows with $t$ |
| Good in | Low frequency | High frequency |

## Key Idea

Use accelerometer for low-frequency reference (gravity is reliable)
Use gyroscope for high-frequency dynamics (no motion artefacts)
Blend them with a **frequency-domain crossover** $\Rightarrow$ Complementary Filter

Kalman filter would adapt the weights dynamically, but the fixed-weight complementary filter is sufficient for our low-speed platform and is far cheaper computationally (1 multiply + 1 add per axis).

# The Complementary Filter

## Discrete recursion

$$\hat{\theta}_k = \alpha\left(\hat{\theta}_{k-1} + \omega_k \Delta t\right) + (1-\alpha)\,\theta_k^{\text{acc}}$$

- $\alpha \approx 0.96$: trust gyroscope short-term, use accelerometer to slowly correct drift.
- Equivalent to: **high-pass on gyro + low-pass on accel**, summing to unity gain.

# Control Problem Statement

## Objective

Drive the platform attitude $(\theta_p, \theta_r)$ to $(0, 0)$ in the presence of disturbances.

Define the error signal:

$$e(t) = \theta_{\text{ref}}(t) - \theta(t) = 0 - \theta(t) = -\theta(t)$$

We want a control law $u(t)$ (servo correction in degrees) that:

- Reacts quickly to disturbances
- Eliminates steady-state error (when the table itself is tilted)
- Avoids overshoot and oscillation

$\Rightarrow$ **PID controller**, the workhorse of feedback control (Week 11).

# The Three Terms of PID

## Continuous form

$$u(t) = \underbrace{K_p\, e(t)}_{\text{Proportional}} + \underbrace{K_i \int_0^t e(\tau)\, d\tau}_{\text{Integral}} + \underbrace{K_d \frac{de(t)}{dt}}_{\text{Derivative}}$$

| Term | Reacts to | Cures | Side effect |
|---|---|---|---|
| P | Current error | Slow response | Steady-state error |
| I | Accumulated error | Steady-state error | Overshoot, slow |
| D | Rate of change of error | Overshoot, damping | Noise amplification |

Analogy:
P = spring (pulls toward target), I = memory (accumulates over time),
D = damper (resists motion)

# Discrete Implementation

Approximate the integral by rectangular sum, derivative by backward difference:

$$\int_0^{t_k} e\,d\tau \approx \sum_{i=0}^{k} e_i \Delta t, \qquad \frac{de}{dt} \approx \frac{e_k - e_{k-1}}{\Delta t}$$

## Discrete PID

$$u_k = K_p\, e_k + K_i \cdot \Delta t \sum_{i=0}^{k} e_i + K_d \cdot \frac{e_k - e_{k-1}}{\Delta t}$$

**Two practical safeguards:**

- **Anti-windup:** clip the integral so it cannot grow unboundedly during saturation
- **Output saturation:** clip $u$ to the servo's physical range

# Discrete Velocity Form: PI Implementation

**Standard Velocity Form PID:** Subtracting positional $u_{k-1}$ from $u_k$ yields the required *change* $\Delta u_k$:

$$\Delta u_k = u_k - u_{k-1} = K_p[e_k - e_{k-1}] + K_i \Delta t\, e_k + \frac{K_d}{\Delta t}[e_k - 2e_{k-1} + e_{k-2}]$$

## Our Velocity Form PI Implementation

To avoid noise amplification from the second difference (D term), our firmware omits the $K_d$ term and implements a **Velocity Form PI** controller:

$$\Delta u_k = K_{P\_vel}\,[e_k - e_{k-1}] + K_{I\_vel}\, e_k$$

**Engineering Insight:** In microcontrollers, the second difference $[e_k - 2e_{k-1} + e_{k-2}]$ is highly sensitive to sensor noise. Dropping it ensures a smoother control signal.

# Velocity Form PI: Mathematical Equivalence

**Why does it work? (When Accumulated)**
Accumulating (integrating) the velocity step $\Delta u_k$ over time mathematically reconstructs the standard positional control law:

$$u_k = \sum_{j=0}^{k} \Delta u_j = K_{P\_vel}\, e_k + K_{I\_vel} \sum_{j=0}^{k} e_j$$

- $K_{P\_vel}$ **acts as Proportional (*P*):**
  Multiplies the first difference $[e_k - e_{k-1}]$. When accumulated, the intermediate terms cancel out (telescoping sum), leaving just the current error $e_k$.
- $K_{I\_vel}$ **acts as Integral (*I*):**
  Multiplies the current error $e_k$ directly. When summed over time, it naturally becomes the accumulated historical error ($\sum e_j$).

# Full Control Pipeline

1. Read IMU, compute $\theta_p$ and $\theta_r$ via complementary filter.
2. Run two independent PIDs:

$$u_p = \mathrm{PID}_p(0 - \theta_p), \quad u_r = \mathrm{PID}_r(0 - \theta_r)$$

3. For each axis, compute the required string lengths using the geometric formulas:

$$\ell_R = \ell_R(u_p), \quad \ell_L = \ell_L(u_p)$$

4. Convert each string length to a servo rotation:

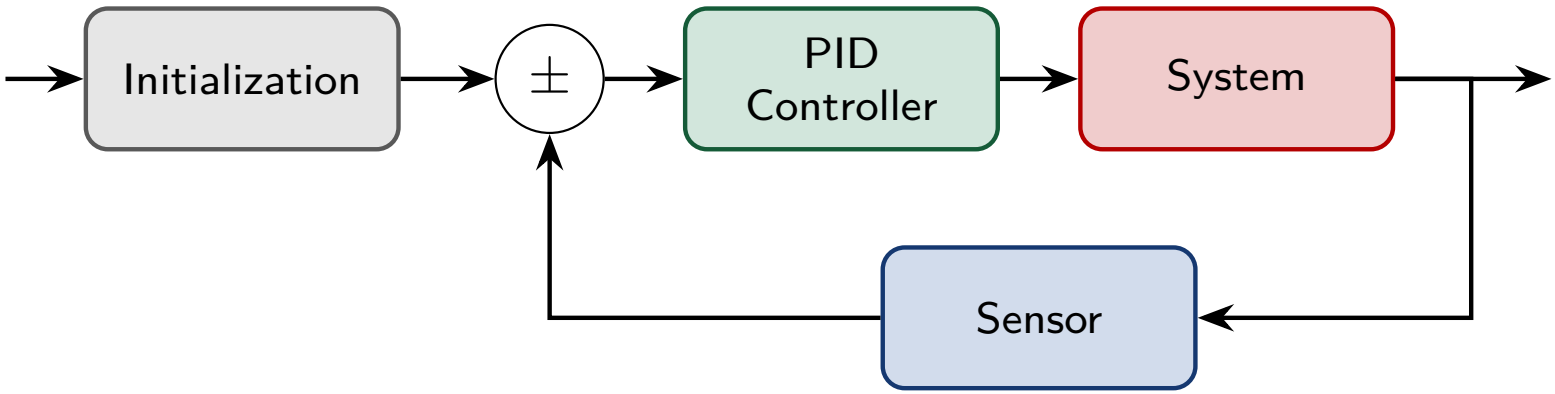
$$\alpha_i = \frac{360^\circ \cdot \big(\ell_i - (a+b)\big)}{C}$$

5. Output PWM via TIM2 channels.

Same procedure runs for the roll axis.
All computations complete within one 2 ms control period.

# Implementation

Overview of our closed-loop control system:



# Implementation

## Code Structure

- **IMU Acquisition (mpu6050.c):**
  - Initialization and calibration
  - Read single frame data
- **Servo Control (servo.c):**
  - Initialization
  - Position-based PWM
- **Attitude Estimation (attitude.c):**
  - Initialization and calibration
  - Complementary filter
- **Control (main.c):**
  - Initialize the servo and IMU modules
  - Core closed-loop control with PI controllers
  - Servo angle calculation

# Implementation: Phase 1

**Initialization**

Three main categories for initializaing:

- **Geometry Bounds:**
  - Upper/Lower pillar length and Servo position
  - Min/Max tilt angles (to prevent over-rotation and string slack)
  - Servo PWM bounds, reversal flags
  - Servo soft deadband and step limit (limit max velocity and ignore small errors)
- **Fixed Control Parameters:**
  - Loop cycle time
  - PID gains $K_{P\_vel}$ and $K_{I\_vel}$
  - Long-term drift compensation
  - Target angle leak gain (to prevent cumulative error)
- **Sensor Calibration:**
  - Accelerometer and gyroscope bias (averaging 500 samples at startup)
  - Complementary filter $\alpha = 0.96$ (favoring gyro instead of accelerometer for stability)
  - Sweep calibration for pitch and roll angles

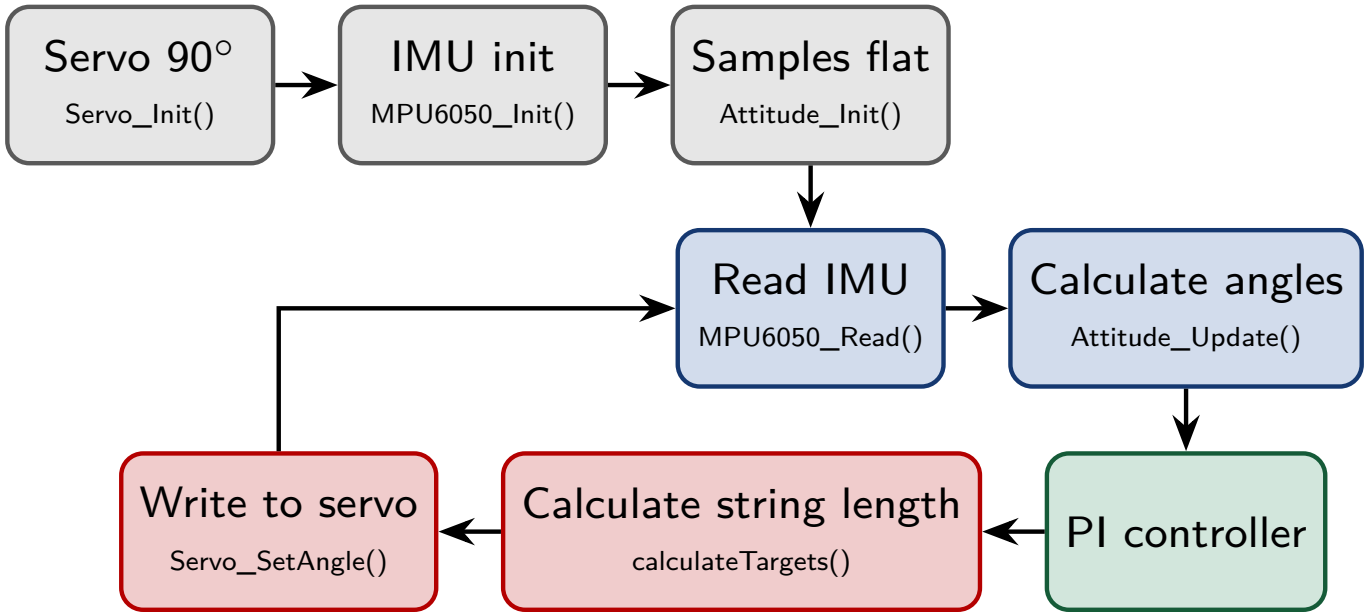
# Implementation: Phase 2

**Closed Loop Control** (Full Control Pipeline)

1. IMU Acquisition: read raw data from MPU6050, then convert to physical units.
2. Attitude Estimation: compute $\theta_p$ and $\theta_r$ with complementary filter.
3. PI Control Calculation: compute $u_p$ and $u_r$
4. String Length Calculation: compute $l_R$ and $l_L$
5. Servo Angle Calculation: compute $\alpha_R$ and $\alpha_L$

# Implementation

## Full Control Pipeline



# Task Model

## Single-rate non-preemptive sequential execution

Control period $T = D = 2000\,\mu$s (500 Hz). Tasks run in fixed order: $\tau_1 \to \tau_2 \to \tau_3 \to \tau_4 \to \tau_5$.

| | **Task** | **Mean ($\mu$s)** | **P99 ($\mu$s)** | **WCET ($\mu$s)** | **Util.** |
|---|---|---|---|---|---|
| $\tau_1$ | IMU Read (I2C) | 1558 | 1558 | 1573 | 78.6% |
| $\tau_2$ | Complementary Filter | 6 | 6 | 7 | 0.4% |
| $\tau_3$ | PID Control | 2 | 3 | 3 | 0.1% |
| $\tau_4$ | Cable Geometry | 31 | 41 | 42 | 2.1% |
| $\tau_5$ | Servo PWM Update | 0 | 2 | 4 | 0.2% |
| | **Total** | 1597 | | **1629** | **81.5%** |

Measured with STM32 DWT cycle counter; 36 031 samples.

# Schedulability Analysis

## Exact test (non-preemptive, single-rate) — necessary & sufficient

$$\sum_i C_i \le D \quad \Longrightarrow \quad 1573 + 7 + 3 + 42 + 4 = 1629 \le 2000 \quad \textbf{SCHEDULABLE}$$

Slack $= D - \sum C_i = 371\,\mu s$ (18.6% margin)

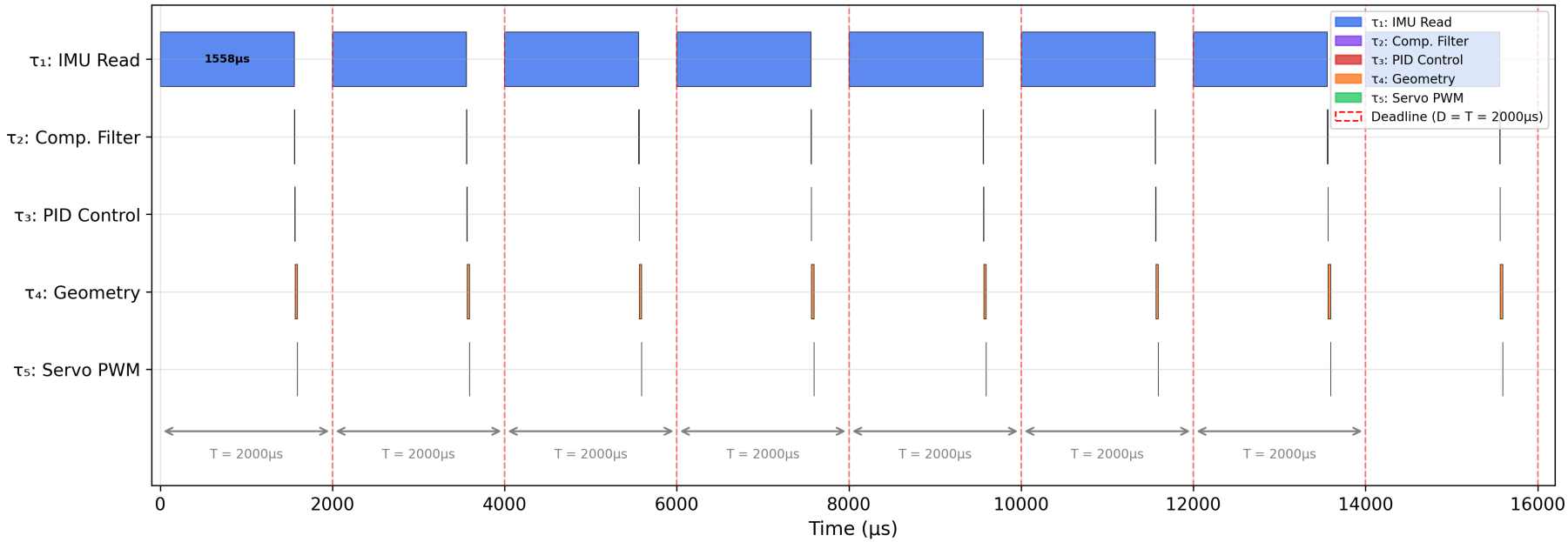
## Liu & Layland RM bound vs. our system

The classical RM utilisation bound for $n = 5$ is $U \le n(2^{1/n} - 1) = 74.3\%$. Our $U = 81.5\%$ exceeds this bound — but that bound applies to *preemptive* RM scheduling. Our system is **non-preemptive single-rate**; the exact test $\sum C_i \le D$ is the correct criterion.

## Empirical verification (36 031 samples)

Deadline misses: **0 / 36 031 (0.00%)** Observed WCET: 1614 $\mu s$ Mean: 1597 $\mu s$ P99: 1608 $\mu s$

# Timing Diagram — Tasks within the Control Period




Each $T$=2000 $\mu$s period runs the five tasks in sequence. $\tau_1$ (IMU read, $\sim$1563 $\mu$s) dominates; $\tau_2$–$\tau_5$ are negligible by comparison. All tasks finish well before the red deadline line $\Rightarrow$ $\sum_i C_i = 1629 \leq D = 2000\,\mu$s, zero misses.

# Lyapunov's Method — Setup

## Lyapunov's direct method (informal)

For equilibrium $\mathbf{x} = \mathbf{0}$, if a scalar $V(\mathbf{x})$ has (i) $V(\mathbf{0})=0$ and $V(\mathbf{x})>0$ for $\mathbf{x}\neq\mathbf{0}$ (positive definite), and (ii) $V$ radially unbounded, then: $\dot{V} \leq 0 \Rightarrow$ **stable / bounded**; $\dot{V} < 0 \Rightarrow$ **asymptotically stable**.

**State:** $\mathbf{x} = [\theta_p,\, \theta_r]^\top$. **Equilibrium:** $\theta_p = \theta_r = 0$ (platform level).

## Lyapunov candidate — quadratic "tilt" function

$$V(\theta_p, \theta_r) = \tfrac{1}{2}\theta_p^2 + \tfrac{1}{2}\theta_r^2 = \tfrac{1}{2}\|\mathbf{x}\|^2$$

Squared distance from level: $V=0$ when flat, larger when tilted. It is a *distance metric* (analogous to spring energy $\frac{1}{2}kx^2$), not literal mechanical energy.

# Continuous-Time Derivative of $V$

**Differentiate by the chain rule** $\left(\frac{d}{dt}\theta^2 = 2\theta\dot{\theta}\right)$:

$$\dot{V} = \frac{d}{dt}\left(\tfrac{1}{2}\theta_p^2 + \tfrac{1}{2}\theta_r^2\right) = \underbrace{\tfrac{1}{2}\cdot 2\theta_p}_{\text{outer}}\underbrace{\dot{\theta}_p}_{\text{inner}} + \tfrac{1}{2}\cdot 2\theta_r\dot{\theta}_r = \boxed{\theta_p\omega_p + \theta_r\omega_r}$$

where $\omega = \dot{\theta}$ is the angular rate (gyro: gy$\rightarrow$pitch, gx$\rightarrow$roll).

**Sign tells us "better or worse" right now:** tilted & returning ($\theta, \omega$ opposite sign) $\Rightarrow \dot{V} < 0$ (improving); tilted & overshooting (same sign) $\Rightarrow \dot{V} > 0$ (worsening).

## Why $\dot{V}$ cannot be signed analytically

Our controller is **velocity-form PI** — the $K_D$ term was dropped, so there is *no explicit rate-damping term*; and we have no identified plant model relating $\omega$ to $\theta$. Hence $\dot{V} = \theta\cdot\omega$ is **not sign-definite** on paper (it is $> 0$ during every overshoot). $\Rightarrow$ we verify the decrease condition **quantitatively from data**, via two measures: the decay rate $\gamma$ and the $\dot{V} \leq 0$ fraction.

# The $\dot{V} \leq 0$ Fraction — What It Measures & Why

We also count, sample by sample, how often the energy is *decreasing*:

$$\rho_{\Delta V} = \frac{1}{M-1} \sum_{k=0}^{M-2} \mathbf{1}\big[\, V[k+1] - V[k] \leq 0 \,\big] \times 100\%$$

where $\mathbf{1}[\cdot]$ is the indicator (1 if true, 0 otherwise). In code: `dV = np.diff(V); frac = np.mean(dV <= 0)*100`.

## Why this equals the Lyapunov decrease test

Since $\dot{V} \approx \dfrac{V[k+1] - V[k]}{\Delta t}$ and $\Delta t > 0$, $\text{sign}(\Delta V) = \text{sign}(\dot{V})$. So $\rho_{\Delta V}$ is the fraction of time the Lyapunov condition $\dot{V} \leq 0$ holds. We check only the *sign* (not $/\Delta t$), i.e. the **direction** of energy change, not its speed.

## Interpreting $\rho_{\Delta V} \approx 50\%$ (our data)

An **under-damped** system oscillates: energy *rises* during overshoot (∼half the time) and *falls* during return (∼half). So $\rho_{\Delta V} \approx 50\%$ is the **signature of oscillation, not instability** — like a swing whose height goes up-down each pass but whose *peaks* shrink. Convergence is judged by the envelope ($\gamma > 0$), not by the per-sample count.

# Decay Rate $\gamma$ — What It Is and Where It Comes From

Beyond $\dot{V} < 0$, the *quantitative* stability condition is that energy decays **proportionally to itself**:

$$\dot{V} \leq -\gamma\, V, \qquad \gamma > 0.$$

**Solving the equality case** (a 1st-order linear ODE):

$$\frac{dV}{dt} = -\gamma V \Rightarrow \frac{dV}{V} = -\gamma\, dt \xrightarrow{\int} \ln V = -\gamma t + C \Rightarrow \boxed{V(t) = V_0\, e^{-\gamma t}}$$

This is **exponential decay** — the same form as a damped spring, an RC discharge, or Newtonian cooling. ($V_0$ = energy at disturbance onset.)

## How we extract $\gamma$ from the log

Take logs: $\ln V(t) = \ln V_0 - \gamma t$ — a straight line. We fit the *peak envelope* of $V$ after each disturbance and read the slope:

$$\gamma = -\,\text{slope of } \ln V_{\text{peak}} \text{ vs } t, \qquad \tau = 1/\gamma \text{ (time constant)}.$$

$\gamma > 0 \Rightarrow V$ decays (converging); $\gamma = 0 \Rightarrow$ no decay; $\gamma < 0 \Rightarrow V$ grows (unstable). **Our result:** $\gamma = 0.275\,\mathbf{s}^{-1}$ **(**$\tau \approx 3.6\,\mathbf{s}$**).**

# Quantitative Stability — Results & Verdict

From 36,031 cycles (80 s), 4 disturbance events:

| Event | $V_{\text{peak}}$ | $\gamma$ ($\text{s}^{-1}$) | $\tau$ (s) | $\rho_{\Delta V}$ |
|---|---|---|---|---|
| #1 | 22.45 | 0.158 | 6.3 | 49.1% |
| #2 | 45.29 | 0.317 | 3.2 | 49.7% |
| #3 | 42.11 | 0.304 | 3.3 | 48.9% |
| #4 | 27.62 | 0.320 | 3.1 | 51.0% |
| **avg** | — | **0.275** | **3.6** | **49.7%** |

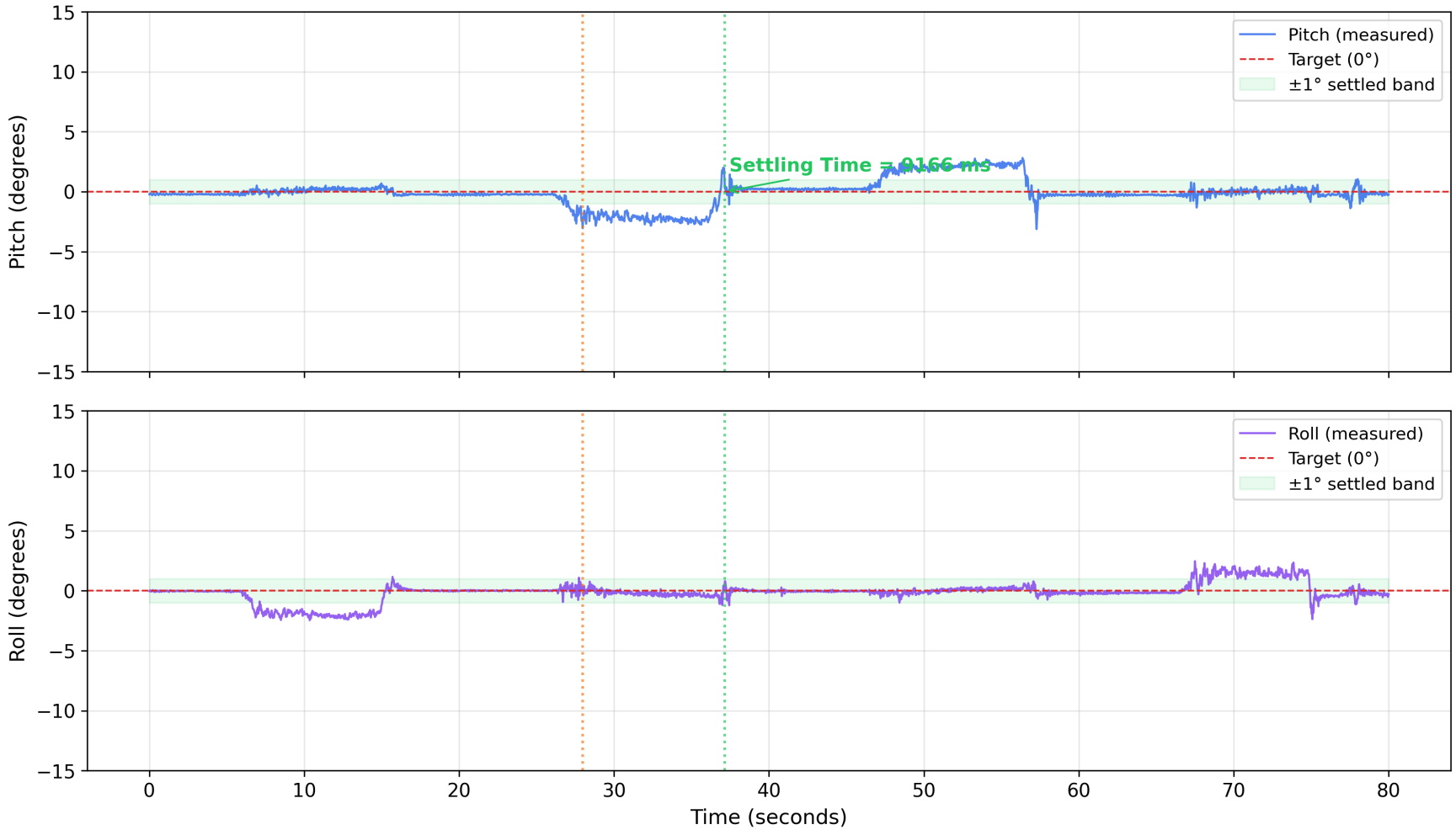
**Three criteria, all met:**

1. **Bounded:** $V_{\max} = 45.3 < \infty$ (no escape).
2. **Convergent:** envelope decays, $\gamma = 0.275\,\text{s}^{-1}$.

## Verdict

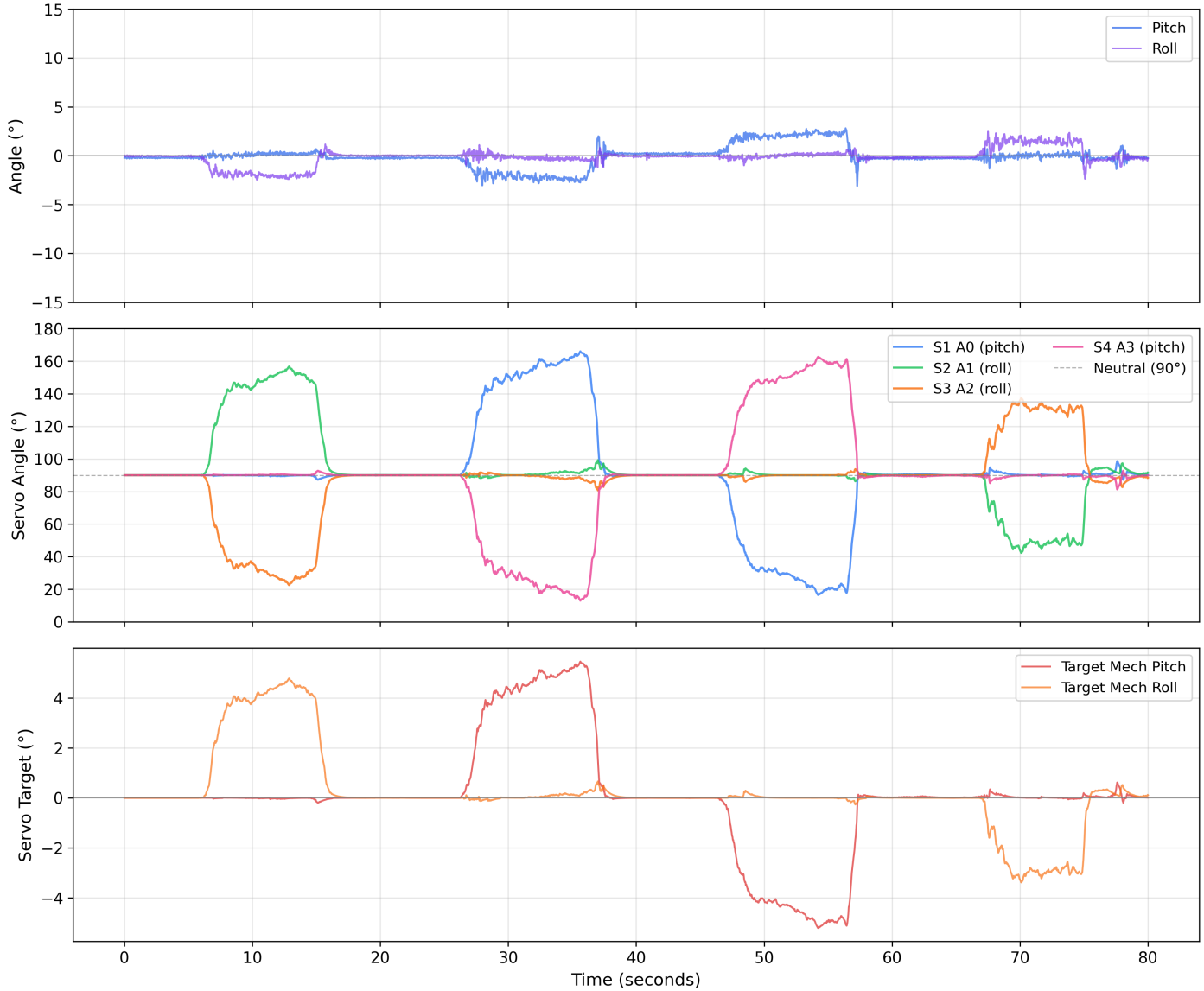
Bounded + decaying envelope + converges to a $\sim 3.4^\circ$ ball $\Rightarrow$ **practically stable / ultimately bounded**. *Not* asymptotic to 0.

# Step Response — Pitch & Roll Axes




Step response (4 disturbance events). Average settling time $\approx 11.9\,\text{s}$ (to $\pm 1^\circ$ band). Peak disturbance $\approx 8.2^\circ$. Platform recovers consistently.

# Control Signal & Servo Behaviour



# Performance Summary

## Real-Time Performance

- Control frequency: **500 Hz** ($T$=2000 $\mu$s)
- WCET: 1629 $\mu$s ($<$ 2000 $\mu$s ✓)
- CPU utilisation: 81.5%
- Deadline margin: 18.6%
- Deadline miss rate: **0.00%**

## Control & Stability

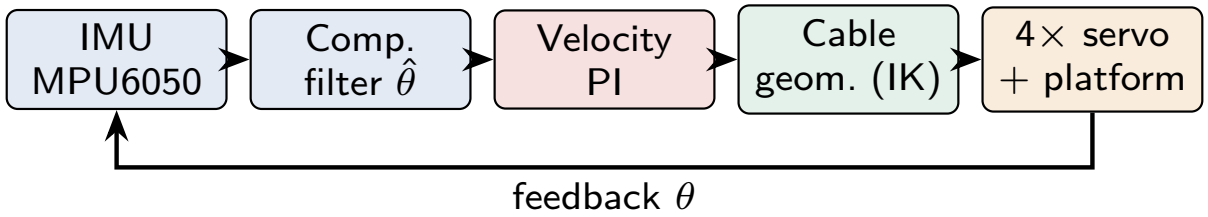
- Avg. settling time: 11.96 s
- Steady-state RMS: 3.38°
- Lyapunov $V$: bounded ($V_{\max}$=45.3)
- Decay rate: $\gamma$=0.275 $s^{-1}$ ✓
- Stability: **practical / ultimately bounded**

## Main bottleneck: IMU I2C (78.6% of budget)

Upgrading to SPI ($\sim$400 $\mu$s) would cut $\tau_1$ WCET by $\sim$75%, freeing $\sim$1600 $\mu$s of slack for a faster loop or a true damping term.

# Project Summary — The Complete CPS Pipeline

One closed loop: **sense** → **estimate** → **control** → **actuate** → **move** → **sense**.




**Each stage**

- Sense/estimate: gyro+accel → drift-free $\hat{\theta}$.
- Control: velocity-form PI ($u$ = integrator) + leak.
- Actuate: IK maps tilt → cable → servo angle.

**Outcomes**

- Real-time: 500 Hz; 1629 $\leq$ 2000 $\mu$s; **0** misses.
- Stability: $V$ bounded, $\gamma$=0.275 $s^{-1}$; ball $\sim 3.4^\circ$.
- **Practically stable / ultimately bounded**.

# Conclusion & Future Work

**What we built and verified:**

- A complete cyber-physical loop: IMU $\rightarrow$ complementary filter $\rightarrow$ velocity-form PI $\rightarrow$ inverse kinematics $\rightarrow$ PWM $\rightarrow$ platform.
- **Formal schedulability**: $\sum C_i = 1629 \leq D = 2000\,\mu$s, 0 deadline misses across 36,031 cycles.
- **Quantitative stability**: Lyapunov $V$ bounded & decaying ($\gamma$=0.275) $\Rightarrow$ practically stable / ultimately bounded.

**Future work (motivated directly by the analysis):**

- **IMU to SPI** ($\rightarrow$ free $\sim$1200 $\mu$s) for a faster loop.
- Systematic $K_P, K_I$ tuning; Kalman filter for attitude.

**Thank you!**

# References

STMicroelectronics.

*RM0383 Reference manual: STM32F411xC/E advanced Arm®-based 32-bit MCUs*.

Rev. 3, STMicroelectronics.

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*MPU-6050 Product Specification*.

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